

Chapter 2

COORDINATE SYSTEMS AND TRANSFORMATION

Education makes a people easy to lead, but difficult to drive; easy to govern but impossible to enslave.

—HENRY P. BROUGHAM

2.1 INTRODUCTION

In general, the physical quantities we shall be dealing with in EM are functions of space and time. In order to describe the spatial variations of the quantities, we must be able to define all points uniquely in space in a suitable manner. This requires using an appropriate coordinate system.

A point or vector can be represented in any curvilinear coordinate system, which may be orthogonal or nonorthogonal.

An orthogonal system is one in which the coordinates are mutually perpendicular.

Nonorthogonal systems are hard to work with and they are of little or no practical use. Examples of orthogonal coordinate systems include the Cartesian (or rectangular), the circular cylindrical, the spherical, the elliptic cylindrical, the parabolic cylindrical, the conical, the prolate spheroidal, the oblate spheroidal, and the ellipsoidal.¹ A considerable amount of work and time may be saved by choosing a coordinate system that best fits a given problem. A hard problem in one coordinate system may turn out to be easy in another system.

In this text, we shall restrict ourselves to the three best-known coordinate systems: the Cartesian, the circular cylindrical, and the spherical. Although we have considered the Cartesian system in Chapter 1, we shall consider it in detail in this chapter. We should bear in mind that the concepts covered in Chapter 1 and demonstrated in Cartesian coordinates are equally applicable to other systems of coordinates. For example, the procedure for

¹For an introductory treatment of these coordinate systems, see M. R. Spiegel, *Mathematical Handbook of Formulas and Tables*. New York: McGraw-Hill, 1968, pp. 124–130.

finding dot or cross product of two vectors in a cylindrical system is the same as that used in the Cartesian system in Chapter 1.

Sometimes, it is necessary to transform points and vectors from one coordinate system to another. The techniques for doing this will be presented and illustrated with examples.

2.2 CARTESIAN COORDINATES (X, Y, Z)

As mentioned in Chapter 1, a point P can be represented as (x, y, z) as illustrated in Figure 1.1. The ranges of the coordinate variables x , y , and z are

$$\begin{aligned} -\infty < x < \infty \\ -\infty < y < \infty \\ -\infty < z < \infty \end{aligned} \quad (2.1)$$

A vector $\mathbf{A}$ in Cartesian (otherwise known as rectangular) coordinates can be written as

$$(A_x, A_y, A_z) \quad \text{or} \quad A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z \quad (2.2)$$

where $\mathbf{a}_x$, $\mathbf{a}_y$, and $\mathbf{a}_z$ are unit vectors along the x -, y -, and z -directions as shown in Figure 1.1.

2.3 CIRCULAR CYLINDRICAL COORDINATES (ρ, ϕ, z)

The circular cylindrical coordinate system is very convenient whenever we are dealing with problems having cylindrical symmetry.

A point P in cylindrical coordinates is represented as (ρ, ϕ, z) and is as shown in Figure 2.1. Observe Figure 2.1 closely and note how we define each space variable: ρ is the radius of the cylinder passing through P or the radial distance from the z -axis; ϕ , called the

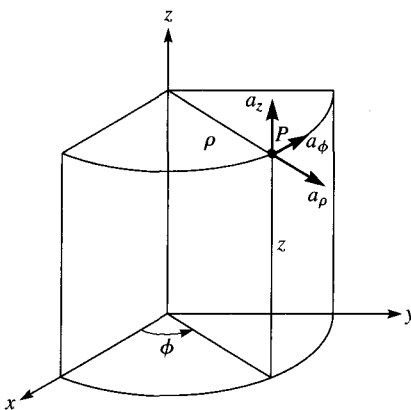


Figure 2.1 Point P and unit vectors in the cylindrical coordinate system.

azimuthal angle, is measured from the x -axis in the xy -plane; and z is the same as in the Cartesian system. The ranges of the variables are

$$\begin{aligned} 0 &\leq \rho < \infty \\ 0 &\leq \phi < 2\pi \\ -\infty &< z < \infty \end{aligned} \quad (2.3)$$

A vector $\mathbf{A}$ in cylindrical coordinates can be written as

$$(A_\rho, A_\phi, A_z) \quad \text{or} \quad A_\rho \mathbf{a}_\rho + A_\phi \mathbf{a}_\phi + A_z \mathbf{a}_z \quad (2.4)$$

where $\mathbf{a}_\rho$, $\mathbf{a}_\phi$, and $\mathbf{a}_z$ are unit vectors in the ρ -, ϕ -, and z -directions as illustrated in Figure 2.1. Note that $\mathbf{a}_\phi$ is not in degrees; it assumes the unit vector of $\mathbf{A}$. For example, if a force of 10 N acts on a particle in a circular motion, the force may be represented as $\mathbf{F} = 10\mathbf{a}_\phi$ N. In this case, $\mathbf{a}_\phi$ is in newtons.

The magnitude of $\mathbf{A}$ is

$$|\mathbf{A}| = (A_\rho^2 + A_\phi^2 + A_z^2)^{1/2} \quad (2.5)$$

Notice that the unit vectors $\mathbf{a}_\rho$, $\mathbf{a}_\phi$, and $\mathbf{a}_z$ are mutually perpendicular because our coordinate system is orthogonal; $\mathbf{a}_\rho$ points in the direction of increasing ρ , $\mathbf{a}_\phi$ in the direction of increasing ϕ , and $\mathbf{a}_z$ in the positive z -direction. Thus,

$$\mathbf{a}_\rho \cdot \mathbf{a}_\rho = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = \mathbf{a}_z \cdot \mathbf{a}_z = 1 \quad (2.6a)$$

$$\mathbf{a}_\rho \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_\rho = 0 \quad (2.6b)$$

$$\mathbf{a}_\rho \times \mathbf{a}_\phi = \mathbf{a}_z \quad (2.6c)$$

$$\mathbf{a}_\phi \times \mathbf{a}_z = \mathbf{a}_\rho \quad (2.6d)$$

$$\mathbf{a}_z \times \mathbf{a}_\rho = \mathbf{a}_\phi \quad (2.6e)$$

where eqs. (2.6c) to (2.6e) are obtained in cyclic permutation (see Figure 1.9).

The relationships between the variables (x, y, z) of the Cartesian coordinate system and those of the cylindrical system (ρ, ϕ, z) are easily obtained from Figure 2.2 as

$$\boxed{\rho = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}, \quad z = z} \quad (2.7)$$

or

$$\boxed{x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z} \quad (2.8)$$

Whereas eq. (2.7) is for transforming a point from Cartesian (x, y, z) to cylindrical (ρ, ϕ, z) coordinates, eq. (2.8) is for $(\rho, \phi, z) \rightarrow (x, y, z)$ transformation.

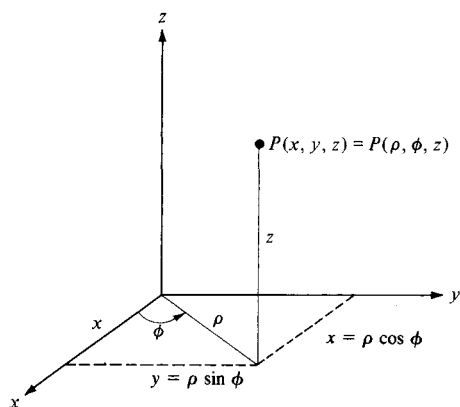


Figure 2.2 Relationship between (x, y, z) and (ρ, ϕ, z) .

The relationships between $(\mathbf{a}_x, \mathbf{a}_y, \mathbf{a}_z)$ and $(\mathbf{a}_\rho, \mathbf{a}_\phi, \mathbf{a}_z)$ are obtained geometrically from Figure 2.3:

$$\begin{aligned}\mathbf{a}_x &= \cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi \\ \mathbf{a}_y &= \sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi \\ \mathbf{a}_z &= \mathbf{a}_z\end{aligned}\quad (2.9)$$

or

$$\begin{aligned}\mathbf{a}_\rho &= \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y \\ \mathbf{a}_\phi &= -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y \\ \mathbf{a}_z &= \mathbf{a}_z\end{aligned}\quad (2.10)$$

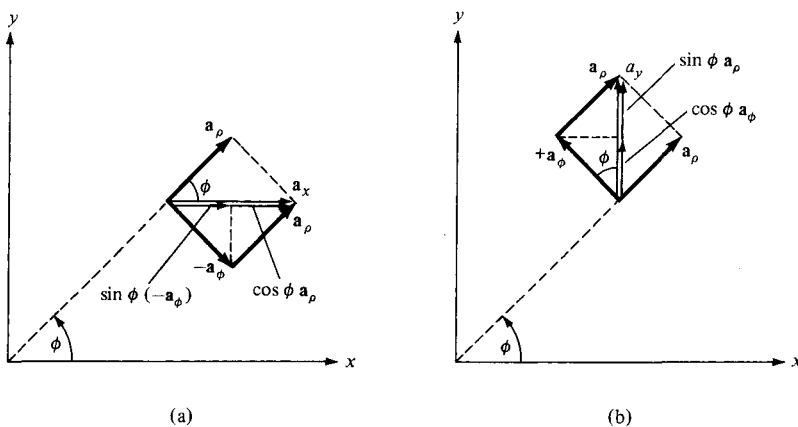


Figure 2.3 Unit vector transformation: (a) cylindrical components of $\mathbf{a}_x$, (b) cylindrical components of $\mathbf{a}_y$.

Finally, the relationships between (A_x, A_y, A_z) and (A_ρ, A_ϕ, A_z) are obtained by simply substituting eq. (2.9) into eq. (2.2) and collecting terms. Thus

$$\mathbf{A} = (A_x \cos \phi + A_y \sin \phi)\mathbf{a}_\rho + (-A_x \sin \phi + A_y \cos \phi)\mathbf{a}_\phi + A_z\mathbf{a}_z \quad (2.11)$$

or

$$\begin{aligned} A_\rho &= A_x \cos \phi + A_y \sin \phi \\ A_\phi &= -A_x \sin \phi + A_y \cos \phi \\ A_z &= A_z \end{aligned} \quad (2.12)$$

In matrix form, we have the transformation of vector $\mathbf{A}$ from (A_x, A_y, A_z) to (A_ρ, A_ϕ, A_z) as

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} \quad (2.13)$$

The inverse of the transformation $(A_\rho, A_\phi, A_z) \rightarrow (A_x, A_y, A_z)$ is obtained as

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} \quad (2.14)$$

or directly from eqs. (2.4) and (2.10). Thus

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} \quad (2.15)$$

An alternative way of obtaining eq. (2.14) or (2.15) is using the dot product. For example:

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \mathbf{a}_x \cdot \mathbf{a}_\rho & \mathbf{a}_x \cdot \mathbf{a}_\phi & \mathbf{a}_x \cdot \mathbf{a}_z \\ \mathbf{a}_y \cdot \mathbf{a}_\rho & \mathbf{a}_y \cdot \mathbf{a}_\phi & \mathbf{a}_y \cdot \mathbf{a}_z \\ \mathbf{a}_z \cdot \mathbf{a}_\rho & \mathbf{a}_z \cdot \mathbf{a}_\phi & \mathbf{a}_z \cdot \mathbf{a}_z \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} \quad (2.16)$$

The derivation of this is left as an exercise.

2.4 SPHERICAL COORDINATES (r, θ, ϕ)

The spherical coordinate system is most appropriate when dealing with problems having a degree of spherical symmetry. A point P can be represented as (r, θ, ϕ) and is illustrated in Figure 2.4. From Figure 2.4, we notice that r is defined as the distance from the origin to